



Intelligence Academy

**Project 01: Image Reconstruction from Top- k
Eigenvectors (PCA/SVD)**

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Top- k Image Reconstruction (Simple Flow)

1.1 Goal in One Line

Given one grayscale image $X \in \mathbb{R}^{12 \times 10}$, find the best rank- k approximation X_k (in Frobenius norm) by centering X , keeping the top k singular layers of the centered matrix, then adding the mean back; report energy, MSE, PSNR (and note SSIM).

1.2 Data Setup (Single 12×10 Image) — Full Mathematical Breakdown

Image dimensions and indexing.

$$H = 12, \quad W = 10, \quad n = HW = 120.$$

$$X \in \mathbb{R}^{12 \times 10}, \quad X_{i,j} = 10i + j, \quad i = 1, \dots, 12, j = 1, \dots, 10.$$

Value range and step sizes.

$$\min X = X_{1,1} = 11, \quad \max X = X_{12,10} = 130.$$

$$X_{i,j+1} - X_{i,j} = 1 \quad (\text{horizontal step}), \quad X_{i+1,j} - X_{i,j} = 10 \quad (\text{vertical step}).$$

$$\text{Arithmetic range midpoint} = \frac{11+130}{2} = \boxed{70.5}.$$

Row-wise arithmetic progressions (AP).

For a fixed i ,

$$X_{i,:} = [10i + 1, 10i + 2, \dots, 10i + 10],$$

$$\text{Row sum: } S_i^{\text{row}} = \sum_{j=1}^{10} (10i + j) = 100i + \underbrace{\sum_{j=1}^{10} j}_{55} = \boxed{100i + 55},$$

$$\text{Row mean: } \mu_i^{\text{row}} = \frac{S_i^{\text{row}}}{10} = 10i + 5.5.$$

$$\text{Row min/max: } \min_j X_{i,j} = 10i + 1, \quad \max_j X_{i,j} = 10i + 10.$$

Column-wise arithmetic progressions (AP).

For a fixed j ,

$$X_{:,j} = [10 \cdot 1 + j, 10 \cdot 2 + j, \dots, 10 \cdot 12 + j],$$

$$\text{Column sum: } S_j^{\text{col}} = \sum_{i=1}^{12} (10i + j) = 10 \underbrace{\sum_{i=1}^{12} i}_{78} + 12j = \boxed{780 + 12j},$$

$$\text{Column mean: } \mu_j^{\text{col}} = \frac{S_j^{\text{col}}}{12} = 65 + j.$$

$$\text{Column min/max: } \min_i X_{i,j} = 10 \cdot 1 + j, \quad \max_i X_{i,j} = 10 \cdot 12 + j.$$

Global sum, mean, and midpoint check.

$$\sum_{i=1}^{12} \sum_{j=1}^{10} X_{i,j} = \sum_{i=1}^{12} \sum_{j=1}^{10} (10i + j) = 10 \cdot W \sum_{i=1}^{12} i + H \sum_{j=1}^{10} j = 10 \cdot 10 \cdot 78 + 12 \cdot 55 = \boxed{8460},$$

$$\bar{x} = \frac{1}{n} \sum_{i,j} X_{i,j} = \frac{8460}{120} = \boxed{70.5} = \frac{\min X + \max X}{2} \quad (\text{as expected for a perfect ramp}).$$

Vectorization maps (useful for PCA/SVD coding).

Column-major (MATLAB/Fortran): $k = (j - 1)H + i$, $x_k = X_{i,j}$, $k = 1, \dots, 120$.

Row-major (C/NumPy default view): $k = (i - 1)W + j$, $x_k = X_{i,j}$, $k = 1, \dots, 120$.

Affine (Kronecker) decomposition.

Let

$$\mathbf{i} = \begin{bmatrix} 1 \\ 2 \\ \vdots \\ 12 \end{bmatrix} \in \mathbb{R}^{12}, \quad \mathbf{j} = \begin{bmatrix} 1 \\ 2 \\ \vdots \\ 10 \end{bmatrix} \in \mathbb{R}^{10}, \quad \mathbf{1}_H \in \mathbb{R}^{12}, \mathbf{1}_W \in \mathbb{R}^{10}.$$

Then

$$X = 10 \mathbf{i} \mathbf{1}_W^\top + \mathbf{1}_H \mathbf{j}^\top,$$

$$\bar{x} = \frac{\mathbf{1}_H^\top X \mathbf{1}_W}{HW} = 10 \cdot \frac{\mathbf{1}_H^\top \mathbf{i}}{H} + \frac{\mathbf{1}_W^\top \mathbf{j}}{W} = 10 \cdot 6.5 + 5.5 = 70.5.$$

Centered coordinates (will be used next).

Define zero-mean coordinate vectors

$$\mathbf{i}_c = \mathbf{i} - 6.5 \mathbf{1}_H = [-5.5, -4.5, \dots, 4.5, 5.5]^\top, \quad \mathbf{j}_c = \mathbf{j} - 5.5 \mathbf{1}_W = [-4.5, -3.5, \dots, 3.5, 4.5]^\top,$$

$$\sum_i (\mathbf{i}_c)_i = 0, \quad \sum_j (\mathbf{j}_c)_j = 0.$$

The globally centered image is

$$\tilde{X} = X - \bar{x} \mathbf{1}_H \mathbf{1}_W^\top = 10 \mathbf{i}_c \mathbf{1}_W^\top + \mathbf{1}_H \mathbf{j}_c^\top.$$

Exact norms and total (for completeness).

$$\|\mathbf{i}_c\|^2 = \sum_{i=1}^{12} (i - 6.5)^2 = \boxed{143}, \quad \|\mathbf{j}_c\|^2 = \sum_{j=1}^{10} (j - 5.5)^2 = \boxed{82.5}.$$

$$\|X\|_F^2 = \sum_{i,j} (10i + j)^2 = \boxed{740,420}, \quad \|\tilde{X}\|_F^2 = \|X\|_F^2 - n\bar{x}^2 = \boxed{143,990}.$$

Per-row / per-column linearity summary.

Along rows: $X_{i,j} = (10i + 5.5) + (j - 5.5)$, slope 1.

Along columns: $X_{i,j} = (65 + j) + 10(i - 6.5)$, slope 10.

Center of mass (in pixel coordinates) = $(i^*, j^*) = (6.5, 5.5)$.

1.3 Centering (Compute $\bar{x}$ and $\tilde{X}$) — Full Mathematical Breakdown

Image size and ramp definition.

$$H = 12, \quad W = 10, \quad n = HW = 120, \quad X_{i,j} = 10i + j \quad (i = 1, \dots, 12; j = 1, \dots, 10).$$

Auxiliary sums (used repeatedly).

$$\sum_{i=1}^{12} i = \frac{12 \cdot 13}{2} = 78, \quad \sum_{i=1}^{12} i^2 = \frac{12 \cdot 13 \cdot 25}{6} = 650,$$

$$\sum_{j=1}^{10} j = \frac{10 \cdot 11}{2} = 55, \quad \sum_{j=1}^{10} j^2 = \frac{10 \cdot 11 \cdot 21}{6} = 385.$$

Global mean $\bar{x}$ (two-line derivation).

$$\sum_{i=1}^{12} \sum_{j=1}^{10} (10i + j) = 10 \underbrace{\left(\sum_{i=1}^{12} i \right)}_{78} \underbrace{\left(\sum_{j=1}^{10} 1 \right)}_{10} + \underbrace{\left(\sum_{i=1}^{12} 1 \right)}_{12} \underbrace{\left(\sum_{j=1}^{10} j \right)}_{55} = 10 \cdot 78 \cdot 10 + 12 \cdot 55 = 8460,$$

$$\bar{x} = \frac{8460}{n} = \frac{8460}{120} = \boxed{70.5} \quad (\text{also } (11 + 130)/2 = 70.5).$$

Global centering and its two-layer decomposition.

Define the zero-mean coordinate vectors

$$\mathbf{i}_c = [1 - 6.5, 2 - 6.5, \dots, 12 - 6.5]^\top = [-5.5, -4.5, \dots, 4.5, 5.5]^\top \in \mathbb{R}^{12},$$

$$\mathbf{j}_c = [1 - 5.5, 2 - 5.5, \dots, 10 - 5.5]^\top = [-4.5, -3.5, \dots, 3.5, 4.5]^\top \in \mathbb{R}^{10},$$

and the all-ones vectors $\mathbf{1}_H \in \mathbb{R}^{12}$, $\mathbf{1}_W \in \mathbb{R}^{10}$. Then

$$\tilde{X} := X - \bar{x} \mathbf{1}_H \mathbf{1}_W^\top = 10 \mathbf{i}_c \mathbf{1}_W^\top + \mathbf{1}_H \mathbf{j}_c^\top.$$

(Each entry: $\tilde{X}_{i,j} = (10i + j) - 70.5 = 10(i - 6.5) + (j - 5.5)$.)

Per-row and per-column checks (first and last rows/columns).

$$\tilde{X}_{1,:} = [11 - 70.5, 12 - 70.5, \dots, 20 - 70.5] = [-59.5, -58.5, \dots, -50.5],$$

$$\tilde{X}_{12,:} = [121 - 70.5, \dots, 130 - 70.5] = [50.5, 51.5, \dots, 59.5],$$

$$\tilde{X}_{:,1} = \begin{bmatrix} 11 - 70.5 \\ 21 - 70.5 \\ \vdots \\ 121 - 70.5 \end{bmatrix} = \begin{bmatrix} -59.5 \\ -49.5 \\ \vdots \\ 50.5 \end{bmatrix}, \quad \tilde{X}_{:,10} = \begin{bmatrix} 20 - 70.5 \\ 30 - 70.5 \\ \vdots \\ 130 - 70.5 \end{bmatrix} = \begin{bmatrix} -50.5 \\ -40.5 \\ \vdots \\ 59.5 \end{bmatrix}.$$

Orthogonality of the two rank-1 layers (Frobenius inner product).

$$\langle 10 \mathbf{i}_c \mathbf{1}_W^\top, \mathbf{1}_H \mathbf{j}_c^\top \rangle = 10 (\mathbf{1}_H^\top \mathbf{i}_c) (\mathbf{1}_W^\top \mathbf{j}_c) = 10 \cdot 0 \cdot 0 = 0.$$

Exact norms of the centered coordinate vectors.

$$\begin{aligned} & \|\mathbf{i}_c\|^2 \\ &= \sum_{i=1}^{12} (i - 6.5)^2 \\ &= \sum i^2 - 13 \sum i + 12 \cdot 6.5^2 \\ &= 650 - 13 \cdot 78 + 12 \cdot 42.25 \end{aligned}$$

$$=650-1014+507=\boxed{143},$$

$$\begin{aligned}\|\mathbf{j}_c\|^2 &= \sum_{j=1}^{10} (j - 5.5)^2 \\ &= \sum j^2 - 11 \sum j + 10 \cdot 5.5^2 \\ &= 385 - 11 \cdot 55 + 10 \cdot 30.25 \\ &= 385 - 605 + 302.5 = \boxed{82.5}.\end{aligned}$$

Total energy $\|X\|_F^2$ (expanded and verified).

$$\|X\|_F^2 = \sum_{i=1}^{12} \sum_{j=1}^{10} (10i + j)^2 = 100 \sum_{i,j} i^2 + 20 \sum_{i,j} ij + \sum_{i,j} j^2,$$

$$100 \sum_{i,j} i^2 = 100 \left(\sum_{i=1}^{12} i^2 \right) \left(\sum_{j=1}^{10} 1 \right) = 100 \cdot 650 \cdot 10 = \boxed{650,000},$$

$$20 \sum_{i,j} ij = 20 \left(\sum_{i=1}^{12} i \right) \left(\sum_{j=1}^{10} j \right) = 20 \cdot 78 \cdot 55 = \boxed{85,800},$$

$$\sum_{i,j} j^2 = \left(\sum_{j=1}^{10} j^2 \right) \left(\sum_{i=1}^{12} 1 \right) = 385 \cdot 12 = \boxed{4,620},$$

$$\Rightarrow \|X\|_F^2 = \boxed{650,000} + \boxed{85,800} + \boxed{4,620} = \boxed{740,420}.$$

Mean (DC) energy and centered energy (Pythagorean split).

$$n \bar{x}^2 = 120 \cdot (70.5)^2 = 120 \cdot 4970.25 = \boxed{596,430},$$

$$\|\tilde{X}\|_F^2 = \|X\|_F^2 - n \bar{x}^2 = \boxed{740,420} - \boxed{596,430} = \boxed{143,990}.$$

Independent recomputation of $\|\tilde{X}\|_F^2$ from the two orthogonal layers.

$$\|10 \mathbf{i}_c \mathbf{1}_W^\top\|_F^2 = 100 \|\mathbf{i}_c\|^2 \|\mathbf{1}_W\|^2 = 100 \cdot \boxed{143} \cdot \boxed{10} = \boxed{143,000},$$

$$\|\mathbf{1}_H \mathbf{j}_c^\top\|_F^2 = \|\mathbf{1}_H\|^2 \|\mathbf{j}_c\|^2 = \boxed{12} \cdot \boxed{82.5} = \boxed{990},$$

$$\Rightarrow \|\tilde{X}\|_F^2 = \boxed{143,000} + \boxed{990} = \boxed{143,990} \quad (\text{matches the split}).$$

Projector form (vectorized centering) and properties.

Let $x = \text{vec}(X) \in \mathbb{R}^{120}$, $\mathbf{1} = \mathbf{1}_{120}$.

$$\mathbf{H} = \mathbf{I} - \frac{1}{120} \mathbf{1} \mathbf{1}^\top, \quad \tilde{x} = \mathbf{H}x, \quad \tilde{X} = \text{unvec}(\tilde{x}).$$

$$\mathbf{H}^\top = \mathbf{H}, \quad \mathbf{H}^2 = \mathbf{H}, \quad \mathbf{H}\mathbf{1} = 0, \quad \|x\|_2^2 = \|\tilde{x}\|_2^2 + 120 \cdot 70.5^2.$$

Summary of exact numeric outcomes.

$$\boxed{\bar{x} = 70.5}, \quad \boxed{\|X\|_F^2 = 740,420}, \quad \boxed{n\bar{x}^2 = 596,430},$$

$$\boxed{\|\tilde{X}\|_F^2 = 143,990}, \quad \boxed{\|\mathbf{i}_c\|^2 = 143}, \quad \boxed{\|\mathbf{j}_c\|^2 = 82.5}.$$

1.4 SVD of $\tilde{X}$ and Energy — Full Mathematical Breakdown

Centered image and its two orthogonal rank-1 layers.

Recall

$$\tilde{X} = 10 \mathbf{i}_c \mathbf{1}_W^\top + \mathbf{1}_H \mathbf{j}_c^\top, \quad \mathbf{i}_c \in \mathbb{R}^{12}, \mathbf{1}_H \in \mathbb{R}^{12}, \mathbf{j}_c \in \mathbb{R}^{10}, \mathbf{1}_W \in \mathbb{R}^{10},$$

with zero sums $\mathbf{1}_H^\top \mathbf{i}_c = 0$ and $\mathbf{1}_W^\top \mathbf{j}_c = 0$.

Exact norms of the building blocks.

$$\|\mathbf{i}_c\|^2 = \boxed{143}, \quad \|\mathbf{j}_c\|^2 = \boxed{82.5}, \quad \|\mathbf{1}_H\|^2 = H = \boxed{12}, \quad \|\mathbf{1}_W\|^2 = W = \boxed{10}.$$

Mutual orthogonality (left and right sides).

Left factors:

$$(10 \mathbf{i}_c)^\top \mathbf{1}_H = 10 \mathbf{1}_H^\top \mathbf{i}_c = 0.$$

Right factors:

$$\mathbf{1}_W^\top \mathbf{j}_c = 0.$$

Hence the two outer products $A_1 := 10 \mathbf{i}_c \mathbf{1}_W^\top$ and $A_2 := \mathbf{1}_H \mathbf{j}_c^\top$ are orthogonal in Frobenius inner product: $\langle A_1, A_2 \rangle = 0$.

Consequences for the SVD (rank and directions).

Because A_1 and A_2 are orthogonal rank-1 matrices with mutually orthogonal left and right directions, the thin SVD of $\tilde{X} = A_1 + A_2$ is obtained by normalizing those directions:

$$u_1 = \frac{10 \mathbf{i}_c}{\|10 \mathbf{i}_c\|} = \frac{\mathbf{i}_c}{\|\mathbf{i}_c\|} \in \mathbb{R}^{12}, \quad v_1 = \frac{\mathbf{1}_W}{\|\mathbf{1}_W\|} = \frac{\mathbf{1}_W}{\sqrt{10}} \in \mathbb{R}^{10},$$

$$u_2 = \frac{\mathbf{1}_H}{\|\mathbf{1}_H\|} = \frac{\mathbf{1}_H}{\sqrt{12}} \in \mathbb{R}^{12}, \quad v_2 = \frac{\mathbf{j}_c}{\|\mathbf{j}_c\|} = \frac{\mathbf{j}_c}{\sqrt{82.5}} \in \mathbb{R}^{10}.$$

The rank is therefore

$$\text{rank}(\tilde{X}) = 2.$$

Exact singular values from outer-product norms.

For a rank-1 matrix $ab^\top$, the (only) nonzero singular value equals $\|a\| \|b\|$. Thus

$$\sigma_1 = \|10 \mathbf{i}_c\| \|\mathbf{1}_W\| = 10\sqrt{143} \cdot \sqrt{10} = \sqrt{143,000},$$

$$\sigma_2 = \|\mathbf{1}_H\| \|\mathbf{j}_c\| = \sqrt{12} \cdot \sqrt{82.5} = \sqrt{990}.$$

Ordering by magnitude gives $\sigma_1 \geq \sigma_2 > 0$. Numerically,

$$\sigma_1 \approx 378.153, \quad \sigma_2 \approx 31.464.$$

Squared singular values and total centered energy.

$$\sigma_1^2 = \boxed{143,000}, \quad \sigma_2^2 = \boxed{990}, \quad \|\tilde{X}\|_F^2 = \sigma_1^2 + \sigma_2^2 = \boxed{143,990}.$$

Explicit SVD reconstruction (centered space).

$$\tilde{X} = \sigma_1 u_1 v_1^\top + \sigma_2 u_2 v_2^\top = \underbrace{10 \mathbf{i}_c \mathbf{1}_W^\top}_{A_1} + \underbrace{\mathbf{1}_H \mathbf{j}_c^\top}_{A_2}.$$

Cumulative energy function (closed forms).

$$\text{CE}(k) = \frac{\sum_{i=1}^k \sigma_i^2}{\sum_{i=1}^2 \sigma_i^2}, \quad \sum_{i=1}^2 \sigma_i^2 = \boxed{143,990}.$$

For $k = 1$:

$$\text{CE}(1) = \frac{\boxed{143,000}}{\boxed{143,990}} = 1 - \frac{\boxed{990}}{\boxed{143,990}} \approx \boxed{0.9931} \text{ (about 99.31\%)}.$$

For $k = 2$:

$$\text{CE}(2) = 1.$$

PCA–SVD eigenvalue check (columns-as-features).

With columns as features, the (biased) column covariance is

$$\mathbf{C}_{\text{cols}} = \frac{1}{H} \tilde{X}^\top \tilde{X} = V \Lambda V^\top, \quad \Lambda = \text{diag}(\lambda_1, \lambda_2),$$

and $\lambda_i = \sigma_i^2 / H$ with $H = 12$. Hence

$$\lambda_1 = \frac{143,000}{12} = \boxed{11,916.\bar{6}}, \quad \lambda_2 = \frac{990}{12} = \boxed{82.5},$$

agreeing with the construction (note λ_2 equals $\|\mathbf{j}_c\|^2$ here).

Key takeaways (all exact).

$$\boxed{\text{rank}(\tilde{X}) = 2}, \quad \boxed{\sigma_1^2 = 143,000}, \quad \boxed{\sigma_2^2 = 990},$$

$$\boxed{\|\tilde{X}\|_F^2 = 143,990}, \quad \boxed{\text{CE}(1) \approx 0.9931}, \quad \boxed{\text{CE}(2) = 1}.$$

1.5 Pick k (Energy or PSNR Rule) — Full Mathematical Breakdown

Given (from the 12×10 centered image $\tilde{X}$).

$$\sigma_1^2 = \boxed{143,000}, \quad \sigma_2^2 = \boxed{990}, \quad \sum_i \sigma_i^2 = \|\tilde{X}\|_F^2 = \boxed{143,990},$$

$$H=12, W=10, n=HW=\boxed{120}.$$

$$\text{For 8-bit metrics we use } L = 255 \Rightarrow L^2 = \boxed{65,025}.$$

Energy rule (cumulative energy).

The cumulative energy up to rank k is

$$CE(k) = \frac{\sum_{i=1}^k \sigma_i^2}{\sum_{i=1}^2 \sigma_i^2} \quad (\text{here rank} = 2).$$

Exact evaluations:

$$CE(0) = 0, \quad CE(1) = \frac{\boxed{143,000}}{\boxed{143,990}} = 1 - \frac{\boxed{990}}{\boxed{143,990}}$$

$\approx \boxed{0.993129 \dots}$ (**99.31%**), $\text{CE}(2) = 1$. *Decisions for typical thresholds α :*

$$\alpha = 0.95 \text{ (95\%)} : \quad \text{CE}(1) = 0.9931 \geq 0.95 \Rightarrow \boxed{k = 1}.$$

$$\alpha = 0.99 \text{ (99\%)} : \quad \text{CE}(1) = 0.9931 \geq 0.99 \Rightarrow \boxed{k = 1}.$$

Thus the energy rule at 95% *and* 99% both select $\boxed{k = 1}$.

PSNR rule (via residual energy).

For truncated rank k , the centered-space residual energy is

$$\mathcal{E}_{\text{res}}(k) = \sum_{i > k} \sigma_i^2.$$

The mean-squared error (equivalently in display space, since mean-shift cancels) is

$$\text{MSE}(k) = \frac{\mathcal{E}_{\text{res}}(k)}{HW} = \frac{\sum_{i>k} \sigma_i^2}{\boxed{120}}.$$

The PSNR (in dB) for a peak L is

$$\text{PSNR}(k) = 10 \log_{10} \left(\frac{L^2}{\text{MSE}(k)} \right) = 10 \log_{10} \left(\frac{\boxed{65,025}}{\frac{\sum_{i>k} \sigma_i^2}{\boxed{120}}} \right).$$

Exact PSNR at $k = 0, 1, 2$.

$$\mathcal{E}_{\text{res}}(0) = \sigma_1^2 + \sigma_2^2 = \boxed{143,990} \Rightarrow \text{MSE}(0) = \frac{143,990}{120} = \boxed{1199.916\bar{6}},$$

$$\text{PSNR}(0) = 10 \log_{10} \left(\frac{65,025}{1199.9166} \right) = 10 \log_{10}(54.187 \dots) \approx \boxed{17.33 \text{ dB}}.$$

$$\mathcal{E}_{\text{res}}(1) = \sigma_2^2 = \boxed{990} \Rightarrow \text{MSE}(1) = \frac{990}{120} = \boxed{8.25},$$

$$\text{PSNR}(1) = 10 \log_{10} \left(\frac{65,025}{8.25} \right) = 10 \log_{10}(7,881.818 \dots) \approx \boxed{38.98 \text{ dB}}.$$

$$\mathcal{E}_{\text{res}}(2) = 0 \Rightarrow \text{MSE}(2) = 0, \quad \text{PSNR}(2) = \boxed{+\infty \text{ dB}}.$$

PSNR target τ (direct inequality to choose k).

A PSNR floor τ (in dB) requires

$$\text{PSNR}(k) \geq \tau \iff \text{MSE}(k) \leq L^2 \cdot 10^{-\tau/10} \iff \sum_{i>k} \sigma_i^2 \leq HW \cdot L^2 \cdot 10^{-\tau/10}.$$

For $\tau = \boxed{40 \text{ dB}}$,

$$\text{RHS} = \boxed{HW} \cdot \boxed{L^2} \cdot 10^{-4} = \boxed{120} \cdot \boxed{65,025} \cdot 10^{-4} = \boxed{780.3}.$$

Check $k = 1$: $\sum_{i>1} \sigma_i^2 = \boxed{990} \not\leq \boxed{780.3} \Rightarrow$ fails 40 dB. Check $k = 2$: $\sum_{i>2} \sigma_i^2 = 0 \leq \boxed{780.3} \Rightarrow$ meets 40 dB.

Decisions (energy vs. PSNR).

Energy thresholds 95%, 99% $\Rightarrow k = 1$.

PSNR floor 40 dB $\Rightarrow k = 2$.

Optional: invert PSNR condition to a residual-energy budget.

For a general target τ ,

$$\mathcal{E}_{\text{res}}(k) \leq HW \cdot L^2 \cdot 10^{-\tau/10}$$

gives an explicit numeric budget that your discarded singular values must not exceed. For our numbers ($H=12$, $W=10$, $L=255$), this budget equals 780.3 at 40 dB, 1236.3 at 38 dB, etc.

1.6 Reconstruct $\tilde{X}_k$ and Add Mean — Full Mathematical Breakdown

Setup (fixed $H=12$, $W=10$, $n=120$).

We use the centered, exactly rank-2 decomposition

$$\tilde{X} = 10 \mathbf{i}_c \mathbf{1}_W^\top + \mathbf{1}_H \mathbf{j}_c^\top,$$

where

$$\mathbf{i}_c = [-5.5, -4.5, \dots, 4.5, 5.5]^\top \in \mathbb{R}^{12}, \quad \|\mathbf{i}_c\|^2 = \boxed{143},$$

$$\mathbf{j}_c = [-4.5, -3.5, \dots, 3.5, 4.5]^\top \in \mathbb{R}^{10}, \quad \|\mathbf{j}_c\|^2 = \boxed{82.5},$$

$$\mathbf{1}_H \in \mathbb{R}^{12}, \quad \|\mathbf{1}_H\|^2 = \boxed{12}, \quad \mathbf{1}_W \in \mathbb{R}^{10}, \quad \|\mathbf{1}_W\|^2 = \boxed{10}.$$

Exact thin SVD ($\tilde{X} = U\Sigma V^\top$).

Choose orthonormal singular vectors

$$u_1 = \frac{\mathbf{i}_c}{\|\mathbf{i}_c\|} = \frac{\mathbf{i}_c}{\sqrt{143}}, \quad v_1 = \frac{\mathbf{1}_W}{\|\mathbf{1}_W\|} = \frac{\mathbf{1}_W}{\sqrt{10}},$$

$$u_2 = \frac{\mathbf{1}_H}{\|\mathbf{1}_H\|} = \frac{\mathbf{1}_H}{\sqrt{12}}, \quad v_2 = \frac{\mathbf{j}_c}{\|\mathbf{j}_c\|} = \frac{\mathbf{j}_c}{\sqrt{82.5}}.$$

The singular values are

$$\sigma_1 = \boxed{\sqrt{143,000}} = 10 \|\mathbf{i}_c\| \|\mathbf{1}_W\|, \quad \sigma_2 = \boxed{\sqrt{990}} = \|\mathbf{1}_H\| \|\mathbf{j}_c\|.$$

Thus the two rank-1 SVD layers are *exactly* the two centered layers:

$$\sigma_1 u_1 v_1^\top = (10 \|\mathbf{i}_c\| \|\mathbf{1}_W\|) \left(\frac{\mathbf{i}_c}{\|\mathbf{i}_c\|} \right) \left(\frac{\mathbf{1}_W^\top}{\|\mathbf{1}_W\|} \right) = \boxed{10 \mathbf{i}_c \mathbf{1}_W^\top},$$

$$\sigma_2 u_2 v_2^\top = (\|\mathbf{1}_H\| \|\mathbf{j}_c\|) \left(\frac{\mathbf{1}_H}{\|\mathbf{1}_H\|} \right) \left(\frac{\mathbf{j}_c^\top}{\|\mathbf{j}_c\|} \right) = \boxed{\mathbf{1}_H \mathbf{j}_c^\top}.$$

Centered rank- k reconstruction.

$$\tilde{X}_k = \sum_{i=1}^k \sigma_i u_i v_i^\top.$$

With the closed forms above, this yields:

$$\tilde{X}_0 = \mathbf{0}_{12 \times 10}, \quad \tilde{X}_1 = \boxed{10 \mathbf{i}_c \mathbf{1}_W^\top}, \quad \tilde{X}_2 = \boxed{10 \mathbf{i}_c \mathbf{1}_W^\top + \mathbf{1}_H \mathbf{j}_c^\top} = \tilde{X}.$$

Entrywise formulas (centered space).

Let $i = 1, \dots, 12$ and $j = 1, \dots, 10$. Because $(\mathbf{i}_c)_i = i - 6.5$ and $(\mathbf{j}_c)_j = j - 5.5$,

$$(\tilde{X}_1)_{i,j} = (10 \mathbf{i}_c \mathbf{1}_W^\top)_{i,j} = 10 (i - 6.5) \cdot 1 = \boxed{10(i - 6.5)},$$

$$(\tilde{X}_2)_{i,j} = 10(i - 6.5) + (j - 5.5) = \boxed{10(i - 6.5) + (j - 5.5)} = \boxed{(10i + j) - 70.5}.$$

Add the mean back to obtain display-space X_k .

We have the global mean

$$\bar{x} = \boxed{70.5}, \quad X_k = \tilde{X}_k + \bar{x} \mathbf{1}_{12 \times 10}.$$

Therefore, entrywise:

$$(X_0)_{i,j} = 0 + 70.5 = \boxed{70.5},$$

$$(X_1)_{i,j} = 10(i - 6.5) + 70.5 = \boxed{10i + 5.5} \quad (\text{row-constant; independent of } j),$$

$$(X_2)_{i,j} = 10(i - 6.5) + (j - 5.5) + 70.5 = \boxed{10i + j} = X_{i,j} \quad (\text{exact}).$$

Concrete numeric slices (to verify).

First row ($i = 1$):

$$X_0[1, :] = [70.5, \dots, 70.5],$$

$$X_1[1, :] = [10 \cdot 1 + 5.5, \dots, 10 \cdot 1 + 5.5] = [\boxed{15.5}, \dots, \boxed{15.5}],$$

$$X_2[1, :] = [11, 12, 13, 14, 15, 16, 17, 18, 19, 20] \quad (\text{exact}).$$

First column ($j = 1$):

$$X_0[:, 1] = \begin{bmatrix} 70.5 \\ \vdots \\ 70.5 \end{bmatrix}, \quad X_1[:, 1] = \begin{bmatrix} 15.5 \\ 25.5 \\ \vdots \\ 125.5 \end{bmatrix} \quad (\text{values } 10i + 5.5),$$

$$X_2[:, 1] = \begin{bmatrix} 11 \\ 21 \\ \vdots \\ 121 \end{bmatrix} \quad (\text{exact}).$$

Residuals & energies (for completeness).

$$\| \tilde{X} - \tilde{X}_0 \|_F^2 = \sigma_1^2 + \sigma_2^2 = \boxed{143,990},$$

$$\| \tilde{X} - \tilde{X}_1 \|_F^2 = \sigma_2^2 = \boxed{990}, \quad \| \tilde{X} - \tilde{X}_2 \|_F^2 = 0.$$

Since adding back $\bar{x}$ cancels in differences,

$$\| X - X_k \|_F^2 = \| \tilde{X} - \tilde{X}_k \|_F^2 \quad (\text{identical MSE/PSNR before clipping}).$$

Summary of special cases.

$$k = 0 : X_0 = \bar{x} \mathbf{1} \text{ (flat image at 70.5);}$$

$$k = 1 : X_1(i, j) = 10i + 5.5 \text{ (row-constant);}$$

$$k = 2 : X_2 = X \text{ (exact).}$$

“latex

1.7 Quality Metrics (MSE, PSNR, SSIM): full step-by-step breakdown for $(H=12, W=10, n=120)$

Given SVD energy of the centered image $\tilde{X}$.

$$\sigma_1^2 = \boxed{143,000}, \quad \sigma_2^2 = \boxed{990}, \quad \|\tilde{X}\|_F^2 = \sigma_1^2 + \sigma_2^2 = \boxed{143,990}.$$

Adding the global mean $\bar{x}$ back does not change differences, so all errors are computed in the centered domain.

Mean-Squared Error (MSE).

For one image with $n = HW = 120$ pixels,

$$\text{MSE}(k) = \frac{\|\tilde{X} - \tilde{X}_k\|_F^2}{HW} = \frac{\sum_{i>k} \sigma_i^2}{120}.$$

$$\text{MSE}(0) = \frac{\sigma_1^2 + \sigma_2^2}{120} = \frac{143,990}{120} = \boxed{1,199.916\bar{6}},$$

$$\text{MSE}(1) = \frac{\sigma_2^2}{120} = \frac{990}{120} = \boxed{8.25},$$

$$\text{MSE}(2) = \frac{0}{120} = \boxed{0}.$$

Peak Signal-to-Noise Ratio (PSNR, in dB).

For 8-bit images, $L = 255$ so $L^2 = 255^2 = \boxed{65,025}$. Then

$$\text{PSNR}(k) = 10 \log_{10} \left(\frac{L^2}{\text{MSE}(k)} \right) = 10 \log_{10} \left(\frac{65,025}{\text{MSE}(k)} \right).$$

$$\text{PSNR}(0) = 10 \log_{10} \left(\frac{65,025}{1,199.9167} \right) = 10 \log_{10}(\boxed{54.1913}) \approx \boxed{17.3393} \text{ dB},$$

$$\begin{aligned} \text{PSNR}(1) &= 10 \log_{10} \left(\frac{65,025}{8.25} \right) = 10 \log_{10} \left(\frac{260,100}{33} \right) \\ &= 10 \log_{10}(\boxed{7,881.8182}) \approx \boxed{38.9663} \text{ dB}, \end{aligned}$$

$$\text{PSNR}(2) = 10 \log_{10} \left(\frac{65,025}{0} \right) = \boxed{+\infty} \text{ dB}.$$

Sanity check (PSNR gain from $k=0$ to $k=1$):

$$\Delta \text{PSNR} = 10 \log_{10} \left(\frac{\text{MSE}(0)}{\text{MSE}(1)} \right) = 10 \log_{10} \left(\frac{143,990}{990} \right) \approx \boxed{21.627} \text{ dB}.$$

SSIM (single-window/global version with standard constants).

Use $K_1 = 0.01$, $K_2 = 0.03$, $L = 255$:

$$C_1 = (K_1 L)^2 = \boxed{6.5025}, \quad C_2 = (K_2 L)^2 = \boxed{58.5225}.$$

Because we add back the *same* global mean, $\mu_X = \mu_{X_k}$, so the luminance term is 1. Let per-pixel (population) variances and covariance be

$$s_X^2 = \frac{1}{n} \|\tilde{X}\|_F^2 = \frac{143,990}{120} = \boxed{1,199.9167}, \quad s_{X_k}^2 = \frac{1}{n} \|\tilde{X}_k\|_F^2, \quad s_{X, X_k} = \frac{1}{n} \langle \tilde{X}, \tilde{X}_k \rangle.$$

Since $\tilde{X}_k$ is the orthogonal projection of $\tilde{X}$, we have $s_{X, X_k} = s_{X_k}^2$. For $k = 1$: $s_{X_1}^2 = \frac{143,000}{120} = \boxed{1,191.6667}$; for $k = 2$: $s_{X_2}^2 = s_X^2$. Thus the global SSIM (with identical means) reduces to

$$\text{SSIM}(k) = \frac{2 s_{X_k}^2 + C_2}{s_X^2 + s_{X_k}^2 + C_2}.$$

$$\text{SSIM}(0) = \frac{2 \cdot 0 + 58.5225}{1,199.9167 + 0 + 58.5225} = \frac{58.5225}{1,258.4392} \approx \boxed{0.0465},$$

$$\text{SSIM}(1) = \frac{2 \cdot 1,191.6667 + 58.5225}{1,199.9167 + 1,191.6667 + 58.5225} = \frac{2,441.8559}{2,450.1060} \approx \boxed{0.9966},$$

$$\text{SSIM}(2) = \frac{2 \cdot 1,199.9167 + 58.5225}{1,199.9167 + 1,199.9167 + 58.5225} = \boxed{1.0000}.$$

Final numeric table (all steps evaluated).

k	$\sum_{i>k} \sigma_i^2$	MSE(k)	PSNR(k) [dB]	SSIM(k)
0	143,990	1,199.9167	17.3393	0.0465
1	990	8.2500	38.9663	0.9966
2	0	0	$+\infty$	1.0000

““

1.8 Final Numbers & Tiny Discussion

Fixed image and sizes (explicit).

$$H = 12, \quad W = 10, \quad n = HW = 120, \quad X_{i,j} = 10i + j \quad (1 \leq i \leq 12, 1 \leq j \leq 10).$$



Global mean (every step shown).

$$\begin{aligned}\bar{x} &= \frac{1}{n} \sum_{i=1}^{12} \sum_{j=1}^{10} (10i + j) \\&= \frac{1}{120} \left(10 \cdot 10 \sum_{i=1}^{12} i + 12 \sum_{j=1}^{10} j \right) \\&= \frac{1}{120} \left(100 \cdot \frac{12 \cdot 13}{2} + 12 \cdot \frac{10 \cdot 11}{2} \right) \\&= \frac{1}{120} (100 \cdot 78 + 12 \cdot 55) = \frac{1}{120} (7,800 + 660) \\&= \frac{8,460}{120} = 70.5 \quad \Rightarrow \quad \boxed{\bar{x} = 70.5}.\end{aligned}$$

Total Frobenius energy $\|X\|_F^2$ (expanded).

$$\begin{aligned}\|X\|_F^2 &= \sum_{i=1}^{12} \sum_{j=1}^{10} (10i + j)^2 \\ &= \underbrace{100 \sum_{i=1}^{12} \sum_{j=1}^{10} i^2}_{(A)} + \underbrace{20 \sum_{i=1}^{12} \sum_{j=1}^{10} ij}_{(B)} + \underbrace{\sum_{i=1}^{12} \sum_{j=1}^{10} j^2}_{(C)}.\end{aligned}$$

$$\sum_{i=1}^{12} i = 78, \quad \sum_{i=1}^{12} i^2 = 650, \quad \sum_{j=1}^{10} j = 55, \quad \sum_{j=1}^{10} j^2 = 385.$$

$$(A) = 100 \cdot \left(\sum_{i=1}^{12} i^2 \right) \cdot W = 100 \cdot 650 \cdot 10 = \boxed{650,000},$$

$$(B) = 20 \cdot \left(\sum_{i=1}^{12} i \right) \left(\sum_{j=1}^{10} j \right) = 20 \cdot 78 \cdot 55 = \boxed{85,800},$$

$$(C) = \left(\sum_{j=1}^{10} j^2 \right) \cdot H = 385 \cdot 12 = \boxed{4,620}.$$

$$\boxed{\|X\|_F^2 = 650,000 + 85,800 + 4,620 = 740,420}.$$

Mean (DC) energy and centered energy (Pythagorean split).

$$n\bar{x}^2 = 120 \cdot 70.5^2 = 120 \cdot 4,970.25 = \boxed{596,430}.$$

$$\|\tilde{X}\|_F^2 = \|X\|_F^2 - n\bar{x}^2 = 740,420 - 596,430 = \boxed{143,990}.$$

Exact centered factorization into two orthogonal rank-1 layers.

$$\tilde{X} = 10 \mathbf{i}_c \mathbf{1}_W^\top + \mathbf{1}_H \mathbf{j}_c^\top,$$

$$\mathbf{i}_c = \begin{bmatrix} -5.5 \\ -4.5 \\ \vdots \\ 4.5 \\ 5.5 \end{bmatrix} \in \mathbb{R}^{12}, \quad \mathbf{j}_c = \begin{bmatrix} -4.5 \\ -3.5 \\ \vdots \\ 3.5 \\ 4.5 \end{bmatrix} \in \mathbb{R}^{10}, \quad \mathbf{1}_H^\top \mathbf{i}_c = 0, \quad \mathbf{1}_W^\top \mathbf{j}_c = 0.$$

$$\|\mathbf{i}_c\|^2 = \sum_{i=1}^{12} (i - 6.5)^2 = 650 - 13 \cdot 78 + 12 \cdot 6.5^2 = 650 - 1,014 + 507 = \boxed{143}.$$

$$\|\mathbf{j}_c\|^2 = \sum_{j=1}^{10} (j - 5.5)^2 = 385 - 11 \cdot 55 + 10 \cdot 5.5^2 = 385 - 605 + 302.5 = \boxed{82.5}.$$

Singular-value energies from orthogonal norms.

$$\sigma_1^2 = \|10 \mathbf{i}_c \mathbf{1}_W^\top\|_F^2 = 100 \|\mathbf{i}_c\|^2 \|\mathbf{1}_W\|^2 = 100 \cdot \boxed{143} \cdot \boxed{10} = \boxed{143,000},$$

$$\sigma_2^2 = \|\mathbf{1}_H \mathbf{j}_c^\top\|_F^2 = \|\mathbf{1}_H\|^2 \|\mathbf{j}_c\|^2 = \boxed{12} \cdot \boxed{82.5} = \boxed{990}.$$

$$\sigma_1^2 + \sigma_2^2 = \boxed{143,000} + \boxed{990} = \boxed{143,990} = \|\tilde{X}\|_F^2.$$

Cumulative energy at $k = 1$ (explicit fraction).

$$\text{CE}(1) = \frac{\sigma_1^2}{\sigma_1^2 + \sigma_2^2} = \frac{143,000}{143,990} = 1 - \frac{990}{143,990} = 1 - \frac{99}{14,399} \approx \boxed{0.9931} \left(\boxed{99.31\%} \right).$$

MSE/PSNR table (8-bit, $L = 255$).

$$\text{MSE}(k) = \frac{\sum_{i>k} \sigma_i^2}{HW}, \quad \text{PSNR}(k) = 10 \log_{10} \left(\frac{255^2}{\text{MSE}(k)} \right).$$

k	$\sum_{i>k} \sigma_i^2$	MSE(k)	PSNR(k) [dB]
0	143,990	$\frac{143,990}{120} = 1,199.9167$	$10 \log_{10} \left(\frac{65,025}{1,199.9167} \right) \approx 17.35$
1	990	$\frac{990}{120} = 8.25$	$10 \log_{10} \left(\frac{65,025}{8.25} \right) \approx 38.97$
2	0	0	$+\infty$

Tiny discussion (decision by requirement).

At $k = 1$: CE $\approx 99.31\%$, PSNR ≈ 38.97 dB (very good).

If you need ≥ 40 dB, take $k = 2$ (exact, zero error).